Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 07
Chapter 13: Statistics

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. The mean of first five prime numbers is: [1]
(A) 5.0 (B) 5.6 (C) 5.8 (D) 6.2

Q2. If $\sum f_i x_i = 1860$ and the mean of the distribution is 31, then the total frequency $\sum f_i$ is: [1]
(A) 50 (B) 55 (C) 60 (D) 65

Q3. In a frequency distribution, the class intervals are $1-10, 11-20, 21-30, 31-40$. The class size is: [1]
(A) 9 (B) 10 (C) 9.5 (D) 10.5

Q4. The median of first 8 composite natural numbers is: [1]
(A) 9 (B) 9.5 (C) 10 (D) 8.5

Q5. If the mean of a data is 27 and mode is 18, then the value of (Mean − Median) is: [1]
(A) 3 (B) 6 (C) 9 (D) 4.5

Q6. **Assertion (A):** For a given distribution, $\sum_{i=1}^{n} f_i(x_i - \bar{x}) = 0$.
Reason (R): The sum of the products of frequencies and deviations of midpoints from the arithmetic mean is always zero. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Convert the following cumulative distribution into standard class intervals: [2]

Marks	Less than 10	Less than 20	Less than 30	Less than 40
Students	5	13	25	32

Q8. Find the class mark and class size of the interval $45-55$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Compute the mode of the following distribution: [3]

Class	0–5	5–10	10–15	15–20	20–25	25–30
Frequency	4	8	18	12	6	2

Q10. If the median of the following data is 35, find the missing frequency k: [3]

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	2	3	k	6	5	3

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

Q11. The median of the following frequency distribution is 525. Find the values of x and y, if the total frequency is 100: [5]

Class Interval	Frequency	Class Interval	Frequency
0–100	2	500–600	20
100–200	5	600–700	y
200–300	x	700–800	9
300–400	12	800–900	7
400–500	17	900–1000	4

Q12. The mean of the following distribution is 62.8 and the sum of all frequencies is 50. Find the missing frequencies f_1 and f_2: [5]

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	5	f_1	10	f_2	7	8

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: INDUSTRIAL MANUFACTURING PLANT DAILY WAGES AND WELFARE AUDIT [4 Marks]

A labor enforcement commissioner conducted a statutory wage survey of 50 skilled and semi-skilled factory workers in an industrial export zone.

Daily Wages (Rs)	200–250	250–300	300–350	350–400	400–450
Number of Workers	12	14	8	6	10

(i) Find the mean daily wage of the factory workers using the Assumed Mean Method. [2]

(ii) Identify the modal wage bracket of the workers. [1]

(iii) Find the number of workers earning less than Rs 350 per day. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★